



Automated detection of arrhythmias using a novel interpretable feature set extracted from 12-lead electrocardiogram

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ABSTRACT

The availability of large-scale electrocardiogram (ECG) databases and advancements in machine learning have facilitated the development of automated diagnostic systems for cardiac arrhythmias. Deep learning models, despite their potential for high accuracy, have had limited clinical adoption due to their inherent lack of interpretability. This study bridges this gap by proposing a feature-based approach that maintains comparable performance to deep learning while providing enhanced interpretability for clinical utility. The method extracts a total of 654 individual features, classified into 60 feature types from each ECG. The features use mathematical techniques such as the Fourier transform, wavelet transform, and cross-correlation for rigorous evaluation of ECG characteristics. The eXtreme Gradient Boosting framework was employed to classify each ECG into one of nine diagnostic classes. Shapley Additive Explanations (SHAP) value analysis was used to downselect the feature set to the minimal set without incurring a performance reduction (159 features). Overall, the proposed method demonstrated performance comparable to state-of-the-art deep learning classifiers, achieving a weighted F1 score of 81% during cross-validation and 68% on the external test dataset, while offering greater ease of implementation and adaptability to diverse clinical applications. Notably, the proposed method demonstrated superior accuracy in identifying atrial fibrillation and block-related abnormalities, achieving overall F1 scores of 89% and 87% during cross-validation and 79% and 75% on the external test dataset, respectively. SHAP value analysis of the testing results revealed the top-performing features for each diagnostic class aligned with standard clinical diagnostic processes, highlighting the clinical interpretability of our approach.

1. Introduction

Cardiovascular diseases (CVDs) are the leading cause of mortality in the world, representing 31% of all global deaths [1]. Early diagnosis of CVDs is crucial for providing timely intervention and preventing severe complications. Arrhythmias are a form of CVD that induce a five-fold increase in the risk of heart attack and stroke. In clinical settings, cardiac arrhythmias are diagnosed using 12-lead electrocardiograms (ECGs), which measure the heart's complete electrical activity and capture the morphological and rhythmic changes caused by arrhythmias. Traditionally, ECGs are read and interpreted manually by clinicians, a process requiring substantial expertise and experience. Unfortunately, this has resulted in misdiagnosis rates of approximately 4 to 47% [2].

Recently, the availability of large-scale ECG databases and advancements in machine learning (ML) methods have paved the way for automated cardiac pathology diagnoses, which can enhance the accuracy and efficiency of identifying arrhythmias [3,4]. Table 1 provides a concise overview of recent studies on automated arrhythmia detection,

highlighting key methodologies, datasets, and their respective strengths and limitations. Early methods primarily focused on extracting relevant features and using appropriate ML classifiers to discriminate between arrhythmias. These extracted features were predominantly based on beat morphology [5,6], ECG rhythm [5,7,8], higher-order statistics [9], Fourier [10] and wavelet transforms [7,8,11], and principal component analysis [6,8]. Commonly used classifiers included Support Vector Machines (SVMs) [11,12] and random forest [13,14]. However, these investigations were often limited by overfitting and low accuracy due to the usage of small ECG datasets (<600 recordings), hindering their clinical applicability.

To address these limitations, many large-scale ECG datasets have been publicly released as part of various challenges, such as the 2018 China Physiological Signal Challenge (2018CPSC, referred to as the CPSC dataset) [1] and 2020 PhysioNet/Computing in Cardiology Challenge (2020CinC, referred to as the PhysioNet dataset) [22]. This has enabled the application of deep learning (DL) techniques for

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Table 1

Summary of feature engineering-based and DL-based methods, including their advantages, shortcomings, and datasets used.

Method	Advantages	Shortcomings	Datasets used
Feature engineering-based method			
Sharmin et al. [15]	High accuracy with limited features; adaptable to clinical settings.	Limited in identifying non-AF abnormalities; lacks instance-specific interpretability.	2017PhysioNet
DL-based Methods			
Lai et al. [16]	Optimal lead selection; captures both local and global patterns.	Lead selection is not arrhythmia-specific; lacks interpretability.	CPSC, PhysioNet
Chen et al. [17]	Identifies arrhythmia-specific lead importance.	Computationally expensive; lacks interpretability.	CPSC
Zhang et al. [18]	Partial interpretability; removes redundant information.	Lacks detailed prediction insights.	CPSC, PTB-XL, Chapman [19]
Yang et al. [20]	Incorporates clinical insights.	Computationally intensive; lacks interpretability.	CPSC, PTB-XL, HFHC [21]

cardiac arrhythmia identification, leveraging advanced ML paradigms such as convolutional neural networks (CNNs) [23,24], long short-term memory networks (LSTMs) [25,26], residual neural networks (ResNets) [27], and transformers [28,29]. For instance, Lai et al. [16] proposed a multi-stage DL model that combined residual CNNs to extract local patterns with bidirectional LSTMs for global patterns and used a forward stepwise subset selection to identify informative ECG leads. The authors observed that the use of optimal lead selection significantly improved the generalizability of the DL model. However, the study employed arrhythmia-independent lead selection, which is suboptimal as the identification of various arrhythmias often depends on lead-specific information. Chen et al. [17] developed a model with five CNN blocks and a bidirectional gated recurrent neural network, enhanced by an attention mechanism and a dense layer, to classify nine ECG types. The study trained models for each ECG lead individually and all 12 leads simultaneously, producing 130 models that were ensembled using genetic algorithm optimization. While this approach, which won first place in the 2018CPSC, effectively identified lead importance for each arrhythmia, it presented significant challenges for implementation due to the complexity of training multiple models and ensembling. Moreover, the random dataset split introduces potential bias, further limiting the model's generalizability.

Although these DL methods demonstrated high accuracy, they offer no insights into the reasoning behind their predictions which limits their clinical utility, as healthcare providers require transparent and explainable systems to make informed decisions. Recent studies have attempted to address this issue by introducing partial interpretability through techniques like attention mechanisms [30] and Gradient-weighted Class Activation Mapping (Grad-CAM) frameworks [31]. He et al. [32] proposed a hybrid DL model that integrated residual networks with bidirectional LSTMs. The model employed Grad-CAM to identify ECG segments contributing the most critical information to its predictions. However, the identification process lacked reliability and was inconsistent in certain cases. Zhang et al. [18] combined transformer encoding and CNNs with self-attention mechanisms to capture multi-scale features, using an information bottleneck to remove redundancy and dual attention mechanisms to address short- and long-term ECG signal aspects. While these attention mechanisms identified important segments for prediction, their reliability and consistency in highlighting relevant segments were limited, reducing their practical effectiveness [33]. Yang et al. [20] proposed a multi-view, multi-scale

deep neural network that fuses leads with similar spatial positions, offering a more clinically relevant approach. However, such efforts remain limited, underscoring the need for further integration of clinical insights into model development. Generally, DL-based arrhythmia detection methods have demonstrated superior performance compared to traditional feature extraction techniques. However, the development of DL models also comes with significant computational costs, requiring extensive training time and powerful hardware such as graphics processing units (GPUs). Additionally, their reliance on raw data often leads to models that are highly dataset-specific, with decreased accuracy when applied to different datasets unless computationally expensive retraining is performed. Consequently, there is a compelling need for ML models that can deliver high accuracy, interpretability, and generalizability to facilitate their integration into clinical practice.

To address this, recent efforts have focused on leveraging novel features with known clinical insights in conjunction with large-scale ECG datasets and advanced ML classifiers to enhance both performance and interpretability [34,35]. Sharmin et al. [15] proposed a novel interpretable feature set combined with eXtreme Gradient Boosting (XGBoost) modeling for atrial fibrillation (AF) detection from Lead 2 only ECGs using the 2017 PhysioNet Atrial Fibrillation Detection Challenge dataset (2017PhysioNet) [36]. This model achieved superior accuracy in distinguishing AF from other categories with a limited feature set. Additionally, it demonstrated adaptability to clinical settings and maintained efficiency even with short-duration ECG recordings. However, a significant limitation of the model was its reduced ability to identify abnormalities other than AF, primarily due to the use of single-lead data. Furthermore, the feature interpretability analysis proposed in the study was limited to evaluating the overall impact of features on predictions across the entire dataset. This approach may fail in cases requiring patient-specific interpretability, where understanding the contributions of features at the level of individual predictions is crucial.

In this work, we develop a single-label, nine-class (“normal sinus rhythm” plus eight arrhythmias) automated ECG diagnostic model that incorporates multi-lead data and patient-specific interpretability, enhancing both the scope and reliability of arrhythmia detection. Clinically relevant features were extracted based on observations of physiological abnormalities, such as irregularities in rhythm, waveform shape, and segment deviations. These abnormalities, initially represented in the time domain across all leads, were transformed into diverse representations in the spectral and modal domains using advanced signal processing and mathematical techniques. To test the efficacy of the features, the XGBoost classifier [37] was employed for its efficiency, robustness, and interpretability when handling structured data. Additionally, the SHapley Additive exPlanations (SHAP) [38] framework was utilized to provide instance-specific interpretability, enabling a deeper understanding of the model's predictions. The 12-lead ECG dataset provided by the 2018CPSC [1] was used to design the method. Hyperparameter tuning of the XGBoost framework was performed and SHAP analysis of the tuned model was used to select a final, trimmed feature set. Then, the “extra” 12-lead ECG dataset provided by the 2018CPSC (CPSC-extra dataset) [22], the Physikalisch Technische Bundesanstalt (PTB-XL dataset) [22,39], and the Georgia 12-lead ECG challenge (G12EC dataset) [22] were used to analyze the accuracy of our model on external datasets for which the model was not trained on. SHAP values were employed to evaluate important features associated with each abnormality. Overall, by combining handcrafted clinically-driven features with XGBoost and leveraging SHAP values to evaluate feature importance, the proposed algorithm aims to take a critical step in bridging the gap between accuracy and interpretability among current automated, multi-class ECG arrhythmia detection algorithms.

The main contributions of this paper are summarized below:

- Utilization of handcrafted interpretable features on large-scale 12-lead ECG datasets for multi-class arrhythmia detection, achieving performance comparable to DL models
- Application of XGBoost to multi-class arrhythmia classification using the extracted features, demonstrating its effectiveness in this domain
- Leveraging SHAP values to analyze the interpretability of extracted features from 12-lead ECG data and identify key features associated with each arrhythmia, thus increasing the clinical utility

The rest of this paper is organized as follows. Section 2 provides a detailed description of the proposed feature extraction techniques and the ML model. Section 3 outlines the datasets used for training and testing, as well as the metrics employed for evaluating the model's performance. Section 4 presents the performance summary of the model, and includes a comparison to benchmark methods. Furthermore, Section 4 includes feature importance analysis and discusses limitations and future directions. Finally, Section 5 concludes the paper.

2. Proposed method

Feature-based ML ECG classification methods aim to identify specific heart arrhythmias by extracting the unique underlying characteristics that distinguish them from other cardiac patterns. Standard clinical practice provides insights into these arrhythmia-specific characteristics. Leveraging this information, feature-based ML classification methods attempt to mimic the clinical process. Consequently, the efficacy of a classifier is heavily influenced by the rigor of its extracted features. In this work, we introduce a novel feature set that is extracted using a combination of signal processing techniques and morphological assessments tailored to each arrhythmia type considered by our method. Fig. 1 provides a visual overview of the proposed methodology, outlining the steps from signal preprocessing to feature extraction and analysis. Initially, the raw ECG signal undergoes noise removal using appropriate signal processing filters. Subsequently, the locations of the PQRST wave morphologies are determined for each heartbeat. A comprehensive analysis of ECG characteristics, including rhythm, segmentation, and time-frequency analyses, is then conducted using tools such as the Fourier transform, wavelet transform, and singular value decomposition. Extracted features are classified as correlation-based metrics, descriptive statistics, and entropy-based measures. These features are then input into an XGBoost-based ECG classifier for arrhythmia evaluation. The model was designed to provide single-class labels among the following nine classes: normal (NORM), AF, first-degree atrioventricular block (IAVB), left bundle branch block (LBBB), right bundle branch block (RBBB), premature atrial contraction (PAC), premature ventricular contraction (PVC), ST-segment depression (STD), and ST-segment elevation (STE). The following sub-sections describe each algorithm step in detail.

2.1. Preprocessing

2.1.1. Signal denoising

Electrode motion artifacts, baseline wander, and powerline interference are three primary noise sources that commonly contaminate ECG signals during acquisition. To address these challenges and improve signal quality, a series of preprocessing steps are applied to each of the 12 ECG leads, individually. To address electrode motion artifacts, a moving median filter with a window size of five sample points was used to identify potential outlier points, with the window size selected to balance the preservation of ECG signal integrity while effectively removing artifacts. Outliers are defined as ECG signal points which deviated more than three median absolute deviations (MAD) from the median of the given window. These identified outlier values are replaced through linear interpolation of non-outlier values using

the built-in MATLAB function *filloutlier*. Next, a high-pass zero-phase Butterworth filter with a cutoff frequency of 0.67 Hz, as recommended by American Heart Association (AHA) guidelines [40], is applied to attenuate baseline wander noise. The powerline interference-based noise is removed using two zero-phase Butterworth notch filters of order 2, corresponding to powerline frequencies of 50 Hz and 60 Hz. A low pass zero-phase Butterworth filter with cutoff frequency of 150 Hz is applied to remove high frequency fluctuations without losing relevant information, as recommended by American National Standards Institute (ANSI) [40]. Finally, a continuous wavelet transform (CWT) and inverse CWT-based denoising was employed to further enhance the signal quality. For this, the CWT coefficients of the ECG signal are calculated using the analytical Morse wavelet. The maximum CWT coefficient magnitude is calculated as the median of the largest 5% of all coefficient magnitude values. CWT coefficient map values less than 10% of the maximum CWT coefficient value are set to zero, while those above this threshold are reduced by the threshold value. The inverse CWT of this thresholded time-frequency representation (TFR) is then computed, resulting in a smooth, denoised ECG signal. This nonlinear filtering process effectively removes noise components not captured by linear methods, while preserving the essential nonlinear signal characteristics. To ensure comparability and consistency across different datasets and individuals, the denoised ECG signals are spectrally normalized to achieve unit spectral energy for each lead.

2.1.2. Morphology detection and beat segmentation

Accurate ECG analysis necessitates precise delineation of the P, Q, R, S, and T wave peaks. Given the QRS complex's prominence as the most distinguishable feature in ECG signals, its accurate identification is crucial for subsequent analysis. Lead 2, offering a clear representation of the QRS complex, was chosen for initial detection. The QRS complexes in Lead 2 are detected using a combination of peak detection algorithms and the Hilbert transform, following the method of Madeiro et al. [41]. Fig. 2 illustrates the identification of the R-peak in the Lead 2 ECG signal. First, the Hilbert transform of the derivative of the Lead 2 ECG signal is calculated (Fig. 2a and b). The squared magnitude of this resulting signal is then computed (referred to here as the "squared Hilbert transform signal"), and peaks in both this signal and the original Lead 2 ECG signal are identified (Fig. 2b). To prevent the detection of multiple peaks within a single QRS complex, a minimum inter-peak distance corresponding to an exceedingly high, physiologically-impossible heart rate (500 beats per minute) is implemented during peak detection. To remove noisy peaks from the squared Hilbert transform signal, a noise threshold was calculated based on the peak magnitude distribution. Specifically, a histogram of the magnitudes of all detected peaks in the squared Hilbert transform signal is computed (Fig. 2c) and the threshold, τ , is determined as:

$$\tau = \frac{\sum_{i=1}^{n_b} pk_i \times N_i}{\sum_{i=1}^{n_b} N_i} \quad (1)$$

where pk_i and N_i represent the peak magnitude and number of peaks corresponding to each histogram bin, respectively, while n_b represents the total number of bins. Peaks with magnitudes below this threshold are considered noise-induced and discarded. For each remaining peak location (which are specific to the squared Hilbert transform signal), the closest prominent peak in the original Lead 2 ECG signal is identified and considered as an R-peak. The prominence of a peak is defined as the minimum signal value difference between the given peak and the lowest point on either side within the interval from the given peak to the adjacent peak. Effectively, this measures the extent to which the given peak stands out due to its height and location relative to the other peaks. Using the R-peak locations, the start and end points of each beat is calculated using the CWT. Specifically, the CWT of the Lead 2 ECG is calculated and the resultant TFR is averaged along the frequency (scale) direction to obtain the temporal variation of the spectral energy of the ECG. The start and end points of each beat

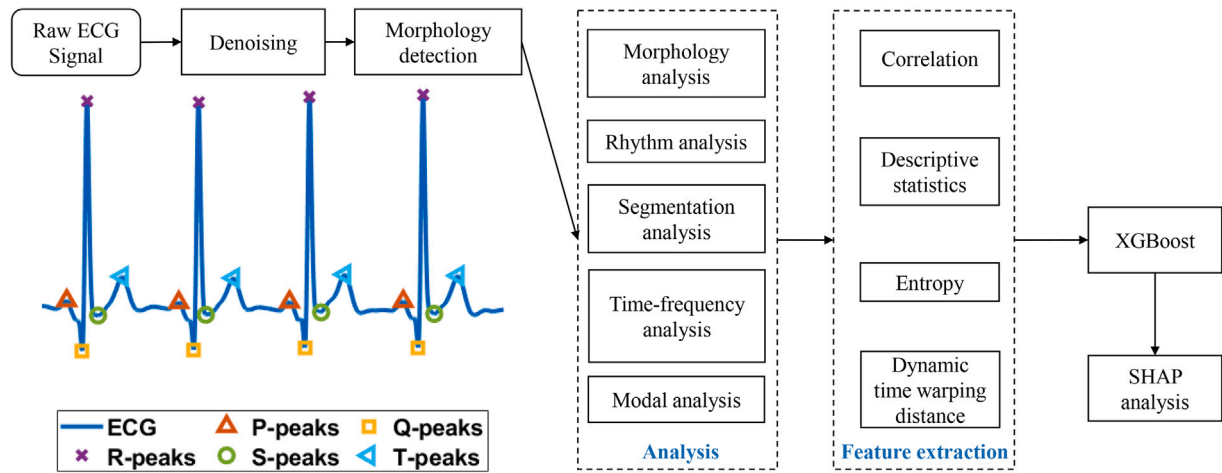


Fig. 1. General schematic of the proposed arrhythmia detection algorithm.

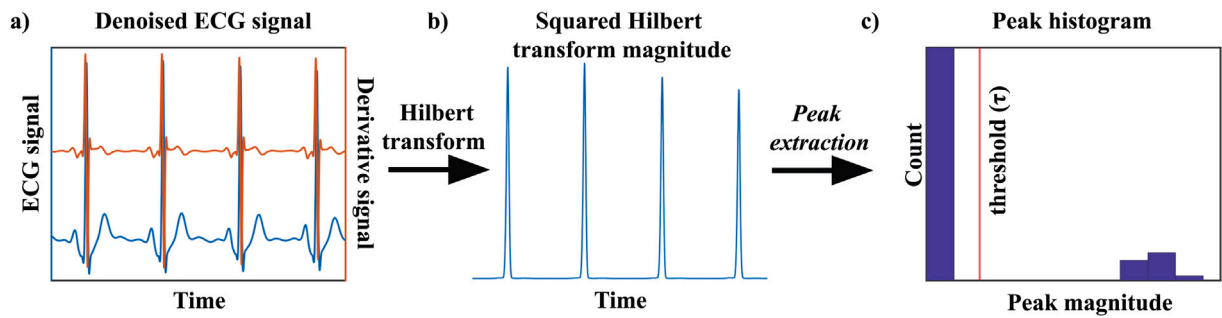


Fig. 2. R-peak extraction approach for use with Lead 2 ECG signals. (a) Denoised Lead 2 ECG signal and its derivative. (b) Squared magnitude of the Hilbert transform of the derivative signal. (c) Histogram of peak magnitudes with noise threshold.

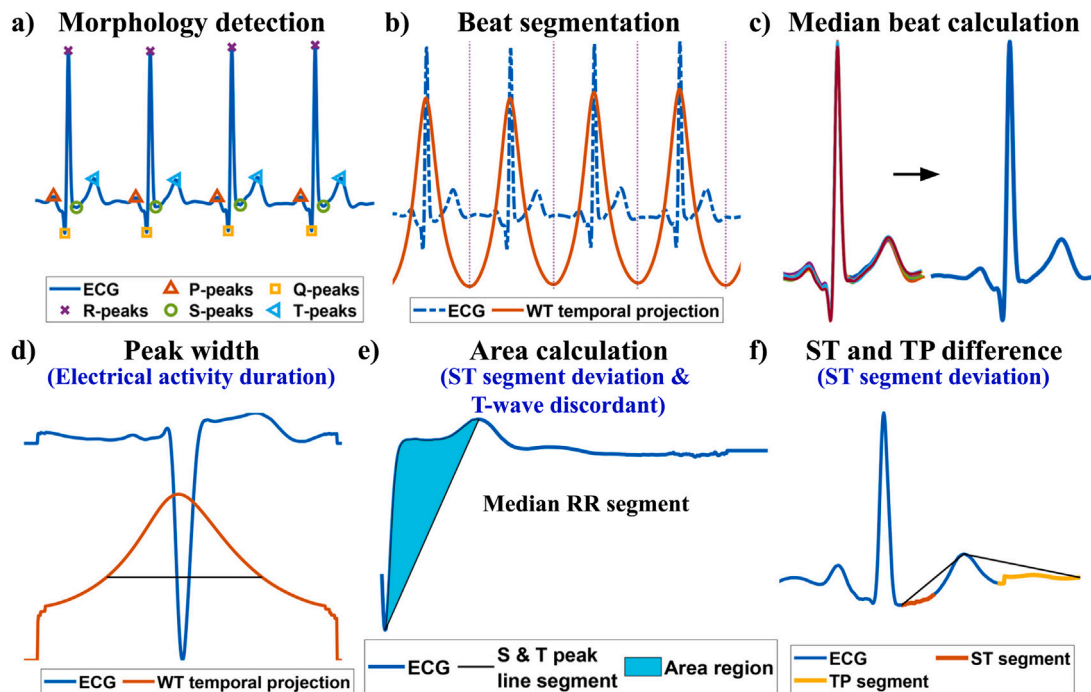


Fig. 3. Proposed feature extraction approach. (a) A sample segment of the processed ECG signal. (b) Beat segmentation of ECG signal. (c) The segmented beat stack and resulting median beat. (d) Calculation of the heart's electrical activity duration using WT temporal projection. (e) The area calculation for ST deviation and T-wave abnormalities. (f) ST segment identification using triangle thresholding.

were defined as the location with the lowest energy between adjacent R-peaks, as illustrated in Fig. 3b. To identify the location of the Q and S-wave peaks, a search interval encompassing normal and abnormal QRS complexes is calculated, following the method of Yeh and Wang [42]. The search interval is calculated assuming a maximum abnormal QRS interval of 220 ms with the R-peak positioned at the center of the window. The most prominent negative peaks of the Lead 2 ECG signal before and after the R-peak inside this search interval are considered as the Q and S-peaks respectively. For the identification of the T-peak, the signal segment from the S-peak to the beat end is extracted from both the positive and negative Lead 2 ECG signals. The peaks within these signal segments are then calculated. The negative Lead 2 ECG signal is considered for this step to account for cases of T-wave inversion, which is commonly associated with various arrhythmias. The most prominent peak within this extracted ECG segment (whether positive or negative) is considered as the T-peak. If the maximum prominence peak was negative, T-wave inversion was assumed to be present. A similar analysis was performed for P-peaks by considering the Lead 2 ECG beat segments from the beat's beginning to the Q-peak. Fig. 3a illustrates the resulting Lead 2 ECG waveform peaks. Finally, the corresponding P, Q, R, S, and T peaks in the remaining 11 ECG leads are computed. For Q, R, and S-peak detection in the other leads, a search interval is defined, extending from 10% of the beat length prior to the Lead 2 Q-peak location to 10% of the beat length after the Lead 2 S-peak location in the same beat. To account for the inverted QRS complex in Leads 4 and 5, appropriate polarity adjustments are applied prior to peak detection. Within this search interval for each beat, the three peaks closest to the corresponding R-peak location in Lead 2 are identified. The most prominent of these three peaks is designated as the R-peak in the given lead. A similar process was applied to identify the Q and S-peaks in the given lead by extracting the peaks of the negative signal within the defined search interval. For the T-peaks, the nearest positive and negative peaks in the given ECG lead signal to the corresponding Lead 2 T-peak locations are identified. The most prominent of the positive and negative peaks is designated as the T-peak for that given lead. This process is repeated to identify the P-peak. Overall, this process results in the complete detection of P, Q, R, S, and T-wave peaks across all 12 leads.

2.2. Feature engineering

A total of 654 features were extracted from the 12-lead ECG signals, corresponding to 60 unique features types. All feature types were analyzed across all 12 leads, except for six feature types related to beat rhythm and beat length, which were not lead-specific. The extracted features can be classified into four categories, namely, rhythm (R), morphology (M), time-frequency (TF) and miscellaneous (Misc). Table 2 provides details of the extracted features, including their descriptions, feature groups, whether they were analyzed across all 12 leads, and the retained leads after feature selection.

2.2.1. Rhythm-based features

The rhythm-based features are predominantly based on the duration of the beats and duration of various ECG segments, e.g., ST-segment. These features represent clinical concepts such as heart rate variability and RR interval volatility. Such features are useful in differentiating rhythm-based heart arrhythmias such as AF, PVC and PAC from other arrhythmias. For the detection of AF, heart rate variability and RR interval volatility are characteristic features. Heart rate variability is calculated as the interquartile range of all beat lengths normalized by the median beat length, while RR interval volatility is defined as the absolute mean of the RR segment deviation, as compared to the previous RR segment. For PAC and PVC, abnormal RR segments should be identified. Thus, for our method, abnormal RR intervals are defined as those deviating by more than three MAD from the median RR interval length. An RR interval with a negative deviation from

the median was classified as a type 1 abnormality, while a positive deviation was classified as a type 2 abnormality. Features such as T-wave to P-wave length and QRS width of each abnormal beat are evaluated. A total of 25 rhythm-based feature types were extracted and are denoted as 'R' in Table 2.

2.2.2. Morphology-based features

The morphology features represent various ECG waveform structural details. Examples of traditional morphology features include the relative ratios of the waveform peak amplitudes and ST segment deviation. Such features are particularly useful for identifying the heart arrhythmias of STE, STD, and RBBB. To evaluate the morphology features, the median beat and median RR interval representing all beats are extracted as illustrated in Fig. 3c. For accurate peak identification of the median beat, the median relative location of the P, Q, R, S, and T-peaks are determined for all beats. Search intervals are established around these locations and the most prominent peak within this search interval, considering peaks of both positive and negative beats, are calculated. The peaks with the highest prominence were then considered as the corresponding P, Q, R, S, and T-wave peaks of the median beat. Several morphology features were developed. Signal analysis techniques, including cross-correlation and dynamic time warping (DTW), are employed to assess the time-lagged and time-wrapped similarity of beats, respectively. For the cross-correlation analysis, extracted features include the mean and standard deviation of the zero-lag cross-correlation values between each beat and the median beat. The mean and standard deviation of the DTW-based distance metric between all beats and the median beat are also calculated, representing the overall similarity and variability of beats with respect to the median beat. Similar analysis was conducted for beat segments from the R-peak to beat's end. Furthermore, the cross correlation and DTW distance metrics are also computed using the adjacent beat rather than the median beat to measure the variability between beats. The positive and negative area between the S to T-peak ECG segments was computed using a linear fit, as shown in Fig. 3e. This area provides a quantitative measure of the ST segment deviation and T-wave abnormalities. Fig. 3f describes the triangle thresholding method used to identify the end of the S and start of the T-wave, representing the ST segments of the median beat. A total of 29 morphology feature types were defined; these features are denoted with an 'M' in Table 2.

2.2.3. Time-frequency and miscellaneous features

The time-frequency and "miscellaneous"-based features all represent metrics developed specifically for our model to innovatively assess arrhythmia-specific characteristics. The CWT provides the temporal distribution of spectral energy resulting in instantaneous spectral analysis. Hence, features corresponding to the CWT provide various measures of non-stationarity in the ECG signal. The TF-based features include entropy of the median WT, as well as the peak magnitude and width of the temporal-projected WT, which respectively indicate the intensity and temporal span of cardiac electrical activity (Fig. 3d). In total, 3 TF-based features types were extracted and denoted as 'TF' in Table 2. The 'miscellaneous features' include all other features used for this study such as spectral-based features, and modal analysis-based features. A total of 3 miscellaneous features were obtained (specified as 'Misc' in Table 2).

3. Materials and methods

We tested the efficacy of the extracted feature set using the XGBoost classifier model on two types of datasets. First, a "development dataset" was employed for model training and cross-validation to establish the baseline performance. Hyperparameter optimization was performed using this dataset, and the resultant optimized model was utilized for a feature selection step using the SHAP framework. The reduced feature space, along with the optimized XGBoost model, was further trained on

Table 2

Extracted feature details, including feature descriptions, groups, 12-lead analysis status, and retained leads after feature selection.

No	Feature description	Group	12-Lead Analysis	Leads (Feature selection)
1	Median beat length	R	No	NA
2	HR variability	R	No	NA
3	Median absolute deviation of beat length	R	No	Removed
4	RR interval volatility	R	No	NA
5	Normalized abnormal beat number (type 1)	R	No	Removed
6	Normalized abnormal beat number (type 2)	R	No	Removed
7	Median PR interval	R	Yes	1, 2, 4, 6, 7, 8, 9, 10, 11, 12
8	Median QT interval	R	Yes	Removed
9	Median ST interval	R	Yes	7
10	Median QRS interval	R	Yes	1, 2, 5, 6, 7, 8, 9
11	Median TP interval	R	Yes	10, 11
12	SD of PR interval	R	Yes	1, 2, 5, 7, 8, 9, 10, 11
13	SD of QT interval	R	Yes	4, 10
14	SD of ST interval	R	Yes	12
15	SD of QRS interval	R	Yes	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
16	SD of TP interval	R	Yes	11
17	Median TP interval of abnormal RR int (type 1)	R	Yes	4
18	SD of TP interval of abnormal RR int (type 1)	R	Yes	Removed
19	Abnormal median QRS width (type 1)	R	Yes	Removed
20	Abnormal QRS width variability (type 1)	R	Yes	Removed
21	Median TP interval of abnormal RR int (type 2)	R	Yes	Removed
22	SD of TP interval of abnormal RR int (type 2)	R	Yes	Removed
23	Abnormal median QRS width (type 2)	R	Yes	Removed
24	Abnormal QRS width variability (type 2)	R	Yes	Removed
25	Median ST segment interval	R	Yes	1, 2, 3, 5, 7, 8, 9, 10, 11, 12
26	Median beat T-R ratio	M	Yes	1, 2, 3, 4, 6, 7, 12
27	Median beat P-R ratio	M	Yes	1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 12
28	Median beat Q-R ratio	M	Yes	1, 2, 4, 5, 6, 7, 9, 10, 11
29	Median beat S-R ratio	M	Yes	1, 2, 4, 6, 7, 8, 10, 12
30	Median beat R-peak width	M	Yes	Removed
31	Median beat Q-peak width	M	Yes	Removed
32	Median beat S-peak width	M	Yes	Removed
33	Median beat T-peak width	M	Yes	Removed
34	Median beat P-peak width	M	Yes	Removed
35	Mean beat correlation with median beat	M	Yes	Removed
36	SD of beat correlation with median beat	M	Yes	1, 9, 10
37	Mean beat correlation with median beat (R to beat end)	M	Yes	Removed
38	SD of beat correlation with median beat (R to beat end)	M	Yes	Removed
39	Positive area: S-T peak line vs. ECG segment	M	Yes	1, 4, 5, 7, 8, 12
40	Negative area: S-T peak line vs. ECG segment	M	Yes	1, 2, 3, 4, 5, 6, 7, 10, 12
41	Mean zero-lag correlation with median beat	M	Yes	8, 9
42	SD of zero-lag correlation with median beat	M	Yes	Removed
43	MAD zero-lag correlation with median beat	M	Yes	Removed
44	Mean zero-lag correlation with adjacent beat	M	Yes	7, 12
45	SD of zero-lag correlation with adjacent beat	M	Yes	Removed
46	MAD zero-lag correlation with adjacent beat	M	Yes	Removed
47	Mean DTW distance with median beat	M	Yes	Removed
48	SD of DTW distance with median beat	M	Yes	Removed
49	Mean DTW distance with adjacent beat	M	Yes	Removed
50	SD of DTW distance with adjacent beat	M	Yes	5
51	Mean correlation abnormal with median RR int (type 1)	M	Yes	Removed
52	Mean correlation abnormal with median RR int (type 2)	M	Yes	Removed
53	Median ST segment deviation	M	Yes	1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12
54	DWT distance between ST and TP interval	M	Yes	2, 3, 5, 7, 8, 10
55	Entropy of median WT	TF	Yes	5, 7, 10, 12
56	Peak magnitude of temporal projected median WT	TF	Yes	1, 3
57	Peak width of temporal projected median WT	TF	Yes	1, 2, 4, 5, 7, 8, 9
58	Spectral spread of median beat	Misc	Yes	1, 4, 5, 7, 8, 9, 10, 12
59	Mean of singular values of beat stack	Misc	Yes	Removed
60	Median of singular values of beat stack	Misc	Yes	3, 4, 6, 7, 9

the entire development dataset. Subsequently, an “external dataset”, completely independent of the training data, was used to assess the model’s generalizability. Fig. 4 illustrates the distribution of classes in both datasets, with Fig. 4a showing the class distribution in the development dataset and Fig. 4b presenting the corresponding distribution in the external dataset. In addition to predicted labels, the model generated prediction probabilities for each class to provide insights into classification confidence. The performance of the proposed model on both datasets was compared to two state-of-the-art DL models. Finally, the feature importance for each arrhythmia class was investigated using SHAP value analysis.

3.1. Dataset description

3.1.1. Development dataset

The publicly-available training dataset from the 2018CPSC served as the development dataset. This dataset contains 6877 12-lead ECG records obtained from 11 hospitals. Each recording was sampled at 500 Hz and had a total recording time between 6 and 60 s. The dataset was included nine diagnostic classes corresponding to the classes our method can identify: NORM, AF, IAVB, LBBB, RBBB, PAC, PVC, STD, and STE. Because herein the proposed method was designed to be a single-label classifier, records containing multiple simultaneous abnormalities (477 cases) were excluded, resulting in a final dataset of 6,400

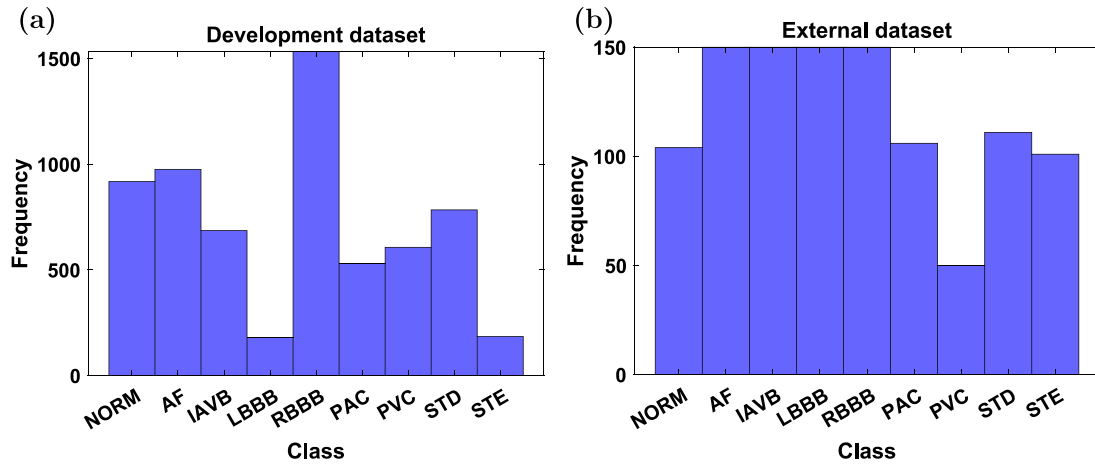


Fig. 4. Distribution of the classes for the (a) development and (b) external datasets.

single-labeled ECGs. For more detailed information about the CPSC dataset, refer to Liu et al. [1]. A stratified sampling of the dataset into 10 folds was employed to ensure a balanced representation of each arrhythmia class during cross-validation using Python's scikit-learn library.

3.1.2. External dataset

The external dataset was compiled from three publicly available sources: the CPSC-extra dataset from the 2018CPSC [22], the PTB-XL dataset provided by Physikalisch Technische Bundesanstalt [22,39], and the G12EC dataset from the Georgia 12-lead ECG Challenge [22]. All datasets were sourced from the 2020CinC [22] and sampled at 500 Hz. For detailed dataset information, readers are referred to Al-day et al. [22]. Because these datasets included both the nine target diagnostic classes and records with additional arrhythmia types, a data filtering process was implemented. Specifically, for each dataset, 12-lead ECG records containing exclusively one of the nine target diagnostic classes were retained, even if other non-target arrhythmia(s) were present. The presence of incomplete or complete LBBB and RBBB was classified as LBBB and RBBB, respectively, regardless of explicit label availability. Both premature contractions (PAC and PVC) and ST-segment changes (STD and STE) were represented by approximately 100 to 150 records across the three datasets. In contrast, NORM instances totaled around 8000 records, primarily from the PTB-XL dataset, resulting in a significantly imbalanced dataset. Hence, to create a balanced test dataset in terms of both class and source dataset, a maximum of 50 records per target arrhythmia class per dataset were extracted. Hence, the resulting final external dataset contained 1072 records with each arrhythmia class comprising between 4.7% and 14.0% of the dataset.

3.2. Evaluation metrics

Six evaluation metrics were employed to assess the classification performance on both the development and external test datasets: accuracy, precision, recall, F1 score, Matthews correlation coefficient (MCC) and area under the receiver operating characteristic curve (AUC-ROC). Accuracy represents the proportion of correct predictions, calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (2)$$

where TP, TN, FP and FN denote true positives, true negatives, false positives, and false negatives, respectively, for a specific diagnostic class. While accuracy provides a measure of the overall performance of the model, it can be misleading when dealing with imbalanced datasets, as it may mask poor performance in minority classes. Since our dataset

is imbalanced, precision and recall metrics were included to provide deeper insights into the classifier's behavior and are calculated by:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

Precision evaluates the proportion of correct positive predictions among all positive predictions, providing insights into the model's tendency to produce false positives, which can lead to costly and unnecessary diagnoses in arrhythmia detection. Recall assesses the classifier's ability to identify all actual positive cases and highlights its effectiveness in minimizing missed diagnoses, which can be potentially life-threatening in the context of arrhythmia detection. To provide a single, balanced measure, the F1-score, the harmonic mean of precision and recall is often employed. The F1-score is calculated as:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

The F1-score is particularly useful for evaluating classifiers on imbalanced datasets, and it is one of the most commonly used metrics in the literature, making it a valuable benchmark for comparing the performance of our model with existing studies. However, the F1-score ignores true negatives, which limits its ability to provide a comprehensive assessment of the classifier's overall performance. To address this limitation and offer a more robust evaluation, MCC is evaluated. MCC assesses the overall quality of the classifier by considering all elements of the confusion matrix (true positives, true negatives, false positives, and false negatives). The MCC is calculated as:

$$\text{MCC} = \frac{(TP \cdot TN) - (FP \cdot FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (6)$$

MCC values range from -1 to $+1$, where $+1$ indicates perfect classification, 0 indicates no better than random guessing, and -1 indicates total disagreement between predictions and true labels. MCC was particularly advantageous for this study, given its ability to account for imbalanced datasets.

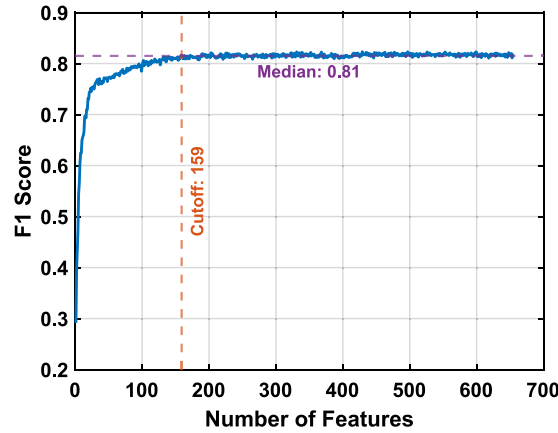
Finally, the AUC-ROC, calculated as the area under the true positive rate (TPR) versus false positive rate (FPR) curve, summarizes the classifier's ability to discriminate between positive and negative classes across different classification thresholds. An ROC curve of a completely random classifier will be the line where the TPR is equal to the FPR, while that of a perfect model will yield a box-type plot passing through the top left corner of the graph and having an AUC of 1.

3.3. XGBoost classifier

XGBoost, an ensemble learning method based on gradient boosting of decision trees, is renowned for its performance on tabular

Hyperparameter	Range	Optimized Value
n_estimators	800 - 1800	1600
max_depth	3 - 11	5
learning_rate	0.01 - 1	0.05
subsample	0.2 - 1.0	0.6
min_child_weight	3 - 9	7
colsample_bytree	0.2 - 1.0	0.8

(a) Complete details of hyperparameter optimization



(b) Variation of median F1 score with the number of top-performing features of the 10-fold cross-validation

Fig. 5. Hyperparameter optimization and feature selection results.

datasets. Given the tabular nature of the extracted features, the XGBoost classifier was employed here. To optimize model performance, a grid search hyperparameter tuning process was conducted using scikit-learn's RandomizedSearchCV. Six key hyperparameters were explored: n_estimators, max_depth, learning_rate, subsample, min_child_weight, and colsample_bytree. The optimization process was performed on the 10-fold cross-validation datasets derived from the development dataset with a split of 9 to 1. To reduce computational cost, 720 hyperparameter combinations were randomly selected using RandomizedSearchCV, and the best-performing combination was chosen as the optimized model. Fig. 5a presents a comprehensive list of all hyperparameters, their corresponding ranges, and final tuned values for the optimized model.

3.4. Feature selection and model interpretation

SHAP values provide a quantitative measure of each feature's impact on specific predictions for a particular diagnostic class, offering insights into feature importance for each arrhythmia type. By calculating SHAP values, we gain a deeper understanding of the contribution of individual features to the overall model predictions and to the classification of each arrhythmia. This provides insight on the decision-making process of the XGBoost model. Here, we employ SHAP value analysis initially as a feature selection tool to reduce the feature space and subsequently as a model interpretation tool to gain deeper insights into the decision-making process of the XGBoost model.

To perform feature selection on the extracted 654 features, SHAP values were calculated for each cross-fold validation of the hyperparameter-tuned model. The SHAP value magnitudes were averaged across all folds, predictions, and classes to determine the relative importance of each feature [43]. Subsequently, cross-fold validation was performed using feature subsets, starting with the most important

feature and progressively adding features in descending order of importance. Fig. 5b shows the median-weighted F1 score as a function of the number of features included, stabilizing at 0.81 when 159 features were included. These selected features, along with their corresponding leads, are detailed in Table 2.

Furthermore, to analyze the decision-making process of the XGBoost model on the external dataset, SHAP values were computed for all features across each prediction for every target arrhythmia. To identify the most influential features for each arrhythmia, mean SHAP value magnitudes [43] were computed across all predictions. These values were then ranked to determine the top features for each class. Additionally, global feature importance was assessed by calculating the average SHAP value magnitude of each feature across all predictions and classes.

4. Results and discussion

4.1. Performance summary

4.1.1. 10-Fold cross-validation

We initially conducted a 10-fold cross-validation on the extracted features using the hyperparameter-tuned XGBoost classifier with a 90/10 training-testing split. Table 3 summarizes the mean performance across all folds for each of the nine diagnostic classes, as well as the overall model performance in terms of precision, recall, and F1-score. Our method achieved macro-averaged performance metrics of 80% precision, 76% recall, and 77% F1-score. However, weighted mean performance metrics, which are based on the number of samples in each diagnostic class, were computed due to the class imbalance evident in Table 3. Our method achieved weighted mean performance values of 82% accuracy, 82% precision, 82% recall, and 81% F1-score. The average MCC score across all folds was calculated to be 0.78. This indicates

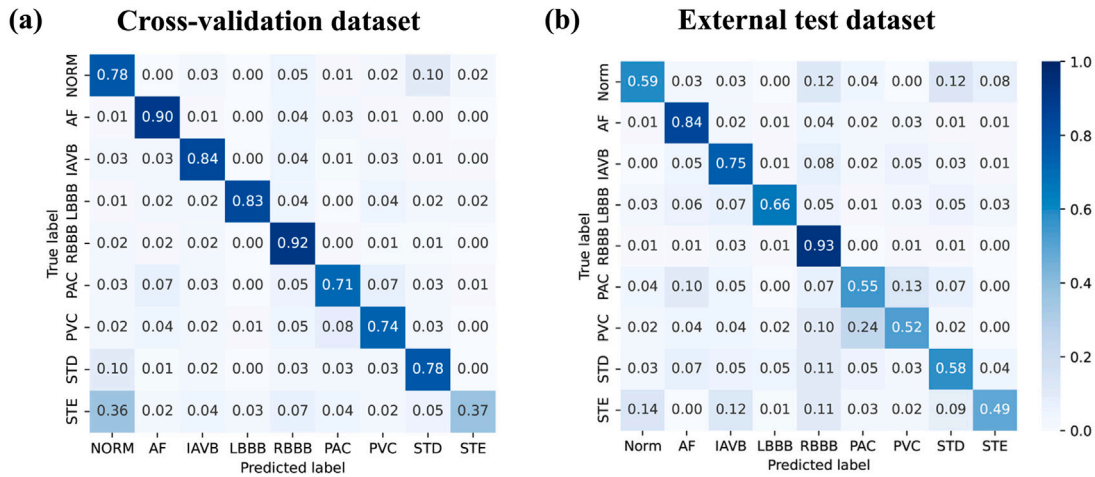


Fig. 6. (a) Normalized confusion matrix of cross-validation averaged across all folds. (b) Normalized confusion matrix of external test dataset prediction.

Table 3

10-Fold cross-validation results for each diagnostic class.

Class	Precision	Recall	F1-Score	Support (%)
NORM	0.76 ± 0.04	0.78 ± 0.05	0.76 ± 0.03	14.37
AF	0.88 ± 0.04	0.90 ± 0.02	0.89 ± 0.02	15.31
IAVB	0.83 ± 0.03	0.84 ± 0.05	0.84 ± 0.03	10.78
LBBB	0.87 ± 0.09	0.83 ± 0.09	0.84 ± 0.06	2.81
RBBB	0.87 ± 0.03	0.92 ± 0.02	0.89 ± 0.02	23.91
PAC	0.74 ± 0.05	0.71 ± 0.06	0.72 ± 0.05	8.28
PVC	0.77 ± 0.05	0.74 ± 0.05	0.75 ± 0.04	9.53
STD	0.79 ± 0.04	0.78 ± 0.06	0.78 ± 0.04	12.19
STE	0.70 ± 0.11	0.37 ± 0.12	0.47 ± 0.11	2.89
Weighted avg	0.82 ± 0.02	0.82 ± 0.02	0.81 ± 0.01	100

a strong overall performance of the model at handling the multi-class classification task, even in the presence of class imbalances. To further evaluate performance, we assessed the model's accuracy on four sub-abnormal groups: AF, blocks (including IAVB, LBBB, and RBBB), premature contractions (including PAC and PVC), and ST-segment changes (including STD and STE). In general, our model demonstrated superior accuracy in identifying AF and the block sub-groups, achieving F1-scores of 89% and 87%, respectively. Within the block sub-group, RBBB exhibited significantly better performance compared to IAVB and LBBB, with improvements of 6% over each. Notably, despite a limited dataset representing only 2.8% of the total data, the model excelled in classifying LBBB, with an F1-score of 84%, highlighting the effectiveness of our feature extraction approach. The premature contraction subgroup demonstrated moderate performance compared to AF and the block sub-group, achieving an F1-score of 74%. This corresponds to a performance reduction of 17% and 15% compared to AF and the block sub-group, respectively. The ST-segment change subgroup exhibited the lowest performance among all categories, with an F1-score of 73%. This was primarily due to the challenges in classifying STE, which achieved an F1-score of only 47%. The detailed analysis of the precision and recall for the STE predictions revealed that the model's accuracy is high when predicting a sample as STE. However, it misses a significant number of actual STE cases, as indicated by the high precision and low recall values. Additionally, the high variability in STE prediction across cross-validation folds suggests insufficient data for robust STE classification. Overall, our model demonstrated strong performance in detecting AF and block patterns, with moderate success in identifying premature contractions and ST-segment changes, with ST-segment changes showing slightly lower performance compared to premature contractions.

Fig. 6a illustrates the normalized confusion matrix of the cross-validation averaged across all 10 folds of our model. Fig. 6a provides

insights into the model's overall performance and potential misclassifications across different classes. In the case of AF and the block sub-abnormal categories, misclassifications were distributed relatively evenly among other classes, reiterating that the current feature set may adequately represent these conditions. The premature contraction subgroup exhibited a tendency to be misclassified as AF, contributing Type-II errors of 7% and 4% for PAC and PVC, respectively. This misclassification likely stems from the model's reliance on rhythm-based features, leading to confusion between these arrhythmias. Additionally, significant overlap was observed between PAC and PVC, with PVC contributing a Type-II error of 7% to PAC classifications and PAC contributing a Type-II error of 8% to PVC classifications. This indicates that the detection of PAC and PVC may over-rely on rhythm-based features rather than morphological differences between the two arrhythmias. Similar to the premature contraction subgroup, the ST-segment change subgroup exhibited significant misclassifications. Specifically, STD was frequently misclassified as NORM, resulting in 10% Type-II errors due to NORM on STD and 10% Type-II errors due to STD on NORM. Furthermore, Fig. 6a shows approximately one-third of STE instances were incorrectly predicted as NORM, resulting in a Type-II error rate of 36%. This significantly impacted recall while maintaining a relatively high precision due to the limited misclassification of NORM as STE. Fig. 6a provides insights into the factors influencing the moderate performance of our model in detecting premature contractions and ST-segment changes. In the case of premature contractions, performance could be improved by identifying and extracting features that effectively differentiate between AF and premature contractions. Furthermore, focusing on morphological features that distinguish between PAC and PVC would be beneficial. Similarly, for the ST-segment category, future analysis should prioritize the extraction of subtle features that differentiate between NORM and ST-segment abnormalities. Improving the accuracy of ST-segment detection across all leads is crucial for enhancing STE classification. Overall, the confusion matrix from cross-validation indicates that our feature set was well-suited for identifying AF and block abnormalities within the context of 12-lead ECG data from the same dataset as the training.

4.1.2. External dataset testing

Next, we investigated the performance of our model on an external test dataset using the hyperparameter-tuned XGBoost classifier trained on the CPSC dataset. Table 4 summarizes the model's performance for the external dataset. Overall, the model achieved weighted average performance metrics of 70% precision, 69% recall, and 68% F1-score on the test dataset. The average MCC score was 0.64. Again, our model demonstrated superior performance in identifying AF and the block sub-group (comprising IAVB, LBBB, and RBBB), achieving F1-scores of

Table 4

Model performance on external test data for each diagnostic class.

Class	Precision	Recall	F1-Score	Support (%)
NORM	0.67	0.59	0.63	9.70
AF	0.75	0.84	0.79	14.00
IAVB	0.71	0.75	0.73	14.00
LBBB	0.90	0.66	0.76	14.00
RBBB	0.66	0.93	0.77	14.00
PAC	0.64	0.55	0.59	9.89
PVC	0.41	0.52	0.46	4.66
STD	0.58	0.58	0.58	10.35
STE	0.71	0.49	0.58	9.42
Weighted Avg	0.70	0.69	0.68	100.00

79% and 75%, respectively. In fact, LBBB demonstrated the highest precision (90%) among all classes, and RBBB exhibited the highest recall (93%) among all classes. The premature contraction and ST-segment change sub-groups displayed moderate-to-low performance, with F1-scores of 54% and 58%, respectively. Similar to the cross-validation results, the model demonstrated high precision (71%) in identifying STE. However, recall remained low at 49%, suggesting missed detection of a significant number of actual STE cases. This improved performance compared to cross-validation might be attributed to the reduced impact of outliers and noise due to the larger test sample size. Overall, the model's performance on the test dataset was 16% lower than that observed during cross-validation, as detailed in Section 4.1.1. Several factors may contribute to this performance discrepancy. The model may have overfitted to the training data, leading to reduced generalization to the unseen test data. Differences in data distribution between the training and test sets could also impact performance. But most importantly, the presence of new patterns, potentially arising from different devices, demographics, or the inclusion of non-target arrhythmias in the test dataset, could challenge the model's ability to accurately classify the target conditions.

Fig. 6b illustrates the normalized confusion matrix of the model predictions of the external test dataset. In general, the model exhibited high detection of actual records (high recall) for both AF and RBBB with Type-II errors of 15% and 7%, respectively. However, a tendency to misclassify NORM, STD, and STE as RBBB was observed, with respective Type-II error rates due to RBBB of 13%, 11%, and 11%. Similarly, STE was frequently misclassified as IAVB, contributing to a Type-II error rate due to IAVB of 12%. While LBBB also experienced some misclassification, the distribution of errors across other classes was relatively even. Similar to the cross-validation results, PAC and PVC exhibited significant overlap, with Type-II error rates of 13% and 24% contributed by PVC and PAC, respectively. Notably, this overlap was more pronounced in the test dataset compared to the cross-validation results, emphasizing the importance of accurately differentiating between PAC and PVC when generalizing the model to new datasets. Furthermore, a significant portion of NORM instances were misclassified as STD, contributing to a Type-II error rate of 13%. Conversely, the misclassification of STD as NORM was less pronounced compared to cross-validation results. The misclassification of STE as NORM decreased to a Type-II error rate of 14%, but this improvement was offset by increased misclassification of STE as RBBB and IAVB. This again highlights the need to identify subtle patterns that differentiate STE from other arrhythmias.

To assess the impact of classification thresholds on the prediction of the nine diagnostic classes, ROC curves were generated for each class using a binarized classification model. Fig. 7 illustrates the ROC curves and corresponding AUC values for each class on the external test dataset. Among all nine diagnostic classes, RBBB exhibited the strongest discriminative performance with an AUC of 0.97. Although PVC demonstrated the lowest AUC of 0.87, it still well surpassed the performance of a random classifier. Generally, the binary classification models for NORM, AF, and the block sub-categories displayed superior

Table 5

Comparison of overall performance on the development dataset and F1 scores for each diagnostic class on the external test dataset.

Development dataset results			
	Our model	Chen et al. [17]	Lai et al. [16]
Reported F1 Score	0.77	0.81	0.76
External dataset results			
Class	Our model	Chen et al. [17]	Lai et al. [16]
NORM	0.63	0.00	0.58
AF	0.79	0.05	0.79
IAVB	0.73	0.08	0.77
LBBB	0.76	0.63	0.75
RBBB	0.77	0.28	0.79
PAC	0.59	0.00	0.51
PVC	0.46	0.24	0.33
STD	0.58	0.02	0.55
STE	0.58	0.15	0.58
Macro Avg	0.65	0.16	0.63

ROC characteristics compared to those for the premature contraction and ST-segment change subcategories, with at least a 4% increase in AUC. The overall mean AUC across all diagnostic classes was found to be 92%, implying satisfactory performance across various classification thresholds.

4.2. Comparison to benchmark

Next, we compared our model's performance to state-of-the-art classifiers available in the literature. Specifically, we selected the CNN-based model proposed by Chen et al. [17] (referred here as CNN model), which was the winning entry in the 2018CPSC, and the multi-stage DL model proposed by Lai et al. [16] (referred here as multi-stage DL model) as benchmarks for comparison. For the cross-validation results, the performance of our XGBoost model was compared to the median performance of the ensemble model reported by Chen et al. [17], and the held-out test dataset performance of the multi-stage model with optimal leads reported by Lai et al. [16]. The score reported by Chen et al. [17] was trained without utilizing the provided multi-label data, while the score reported by Lai et al. [16] utilized the multilabel data. For the external test data performance, model files provided by both Chen et al. [17] and Lai et al. [16] were utilized. Table 5 summarizes the performance of each model for each dataset. In the cross-fold validation on the development dataset, the CNN model demonstrated superior performance compared to both our model and the multi-stage DL model. However, the overall performance reduction for our model was relatively modest, with a difference of only 4%. Additionally, while our model showed better performance than the multi-stage DL model in this comparison, it should be noted that the multi-stage DL model evaluates multilabel data, making direct comparisons challenging. The superior performance of the CNN model is largely due to its significantly better identification of PAC and PVC. However, it also faced challenges in accurately identifying STE, highlighting the inherent difficulty of diagnosing STE due to its subtle and complex diagnostic criteria.

On the external test dataset, our model outperformed both the CNN model and the multi-stage DL model, achieving a 48% and 2% improvement in overall performance, respectively. To validate the statistical significance of these improvements, we repeated our model's performance evaluation using five different random seeds, and a one-sample t-test yielded p-values of <0.0001 and 0.0001 when compared to the fixed benchmark performances of the CNN model and the multi-stage DL model, respectively. While the multi-stage DL model demonstrated a 5% higher performance in identifying IAVB, it significantly underperformed in predicting NORM, PAC, and PVC, with our model achieving performance improvements of 5%, 8%, and 13%, respectively. The

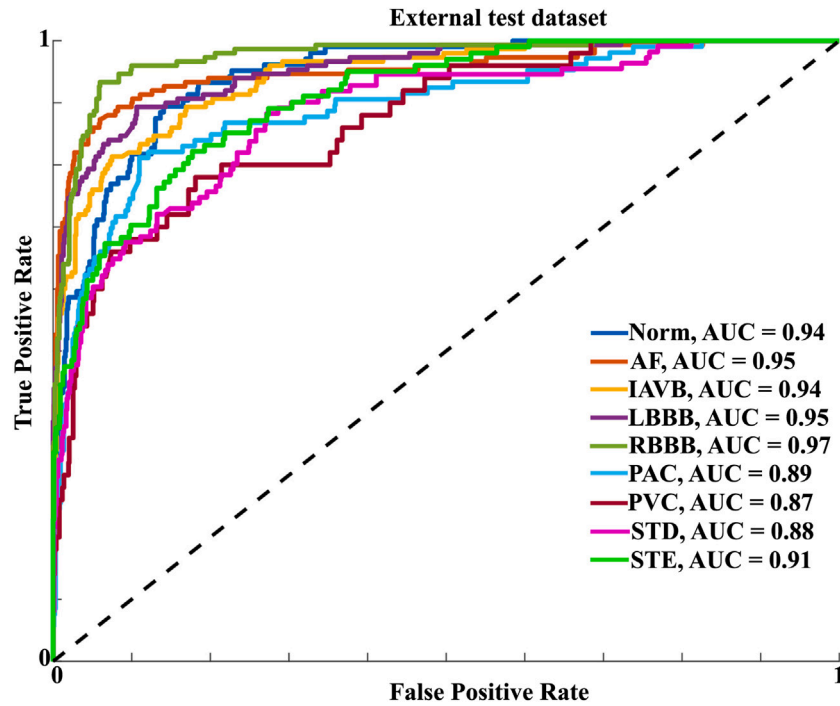


Fig. 7. ROC curve and AUC for nine diagnostic class for external test data prediction.

superior accuracy of our model in recognizing NORM, PAC, and PVC contributed substantially to its higher overall performance. Fig. 8 illustrates sample Lead 2 ECG recordings for various cases of IAVB, PVC, and PAC. Fig. 8a represents a case where our model incorrectly predicted the outcome, while the multi-stage DL model correctly identified the arrhythmia. In this case, the less distinguishable morphology in the ECG caused our model to misinterpret the features. Conversely, the multi-stage DL model correctly identified the arrhythmia, as it does not rely on explicit morphology identification for its predictions. In contrast, Fig. 8b to 8d depict cases where our model correctly predicted the arrhythmia, while the multi-stage DL model made incorrect predictions. In Fig. 8b, the presence of a clear arrhythmia-induced morphology change allowed our model to accurately identify IAVB, whereas the multi-stage DL model struggled due to its reliance on raw signals, making it susceptible to noise. This issue is particularly observed in Fig. 8c (PAC) and Fig. 8d (PVC), where significant baseline wander in the raw signal confused the multi-stage DL model, leading it to misinterpret the variations as ST-segment changes. In contrast, our model, equipped with proper preprocessing steps, effectively handled the noise and accurately identified the arrhythmic beats. While the CNN model demonstrated some capability in identifying LBBB, it completely failed to recognize NORM, PVC, and STD classes. This substantial performance gap can be attributed to training on raw data, as the model files provided by Chen et al. [17] were trained in a different setting from the external test dataset. This limits the CNN model's adaptability to new (unseen) data characterized by variations in devices and demographics. In theory, the performance of both models could be improved by retraining them on normalized or denoised signals. However, such an approach requires extensive computational resources, making it impractical in many settings. Furthermore, once trained, any modifications—such as applying different normalization techniques or addressing additional types of noise—necessitate retraining the model. This underscores a common challenge with DL-based methods: their limited generalizability to diverse clinical settings without significant retraining and reliance on specialized hardware like GPUs. These requirements present substantial barriers to practical implementation, particularly in resource-constrained environments. In contrast, our feature extraction-based approach, while achieving comparable accuracy

Table 6

Important features of each arrhythmia.

Class	1st most important feature	2nd most important feature
Norm	SD of TP interval (lead 11)	Median beat T-R ratio (lead 4)
AF	Heart rate variability	RR interval volatility
IAVB	Median PR interval (lead 10)	Median PR interval (lead 1)
LBBB	Spectral spread of median beat (lead 8)	Peak width of temporal projected median WT (lead 4)
RBBB	Negative area: S-T peak line vs. ECG segment (lead 7)	Positive area: S-T peak line vs. ECG segment (lead 7)
PAC	RR interval volatility	Median TP interval of abnormal RR int (type 1) (lead 4)
PVC	Peak magnitude of temporal projected median WT (lead 3)	SD of beat correlation with median beat (lead 10)
STD	RR interval volatility	Median ST segment interval (lead 10)
STE	Median TP interval (lead 11)	Median ST segment interval (lead 10)

to deep learning methods, offers distinct advantages in terms of training efficiency, interpretability, and adaptability to new data, making it a more suitable choice for clinical implementation.

4.3. Feature interpretability analysis

We next investigated the global and local interpretability of the extracted feature set using SHAP value analysis applied to the external test dataset predictions. Table 6 summarizes the top two most important features for each diagnostic class during the prediction of the external test dataset. The top features identified through SHAP value analysis often aligned well with standard clinical ECG diagnostic practices. For instance, heart rate variability and RR interval volatility were found to be crucial features for AF diagnosis, consistent with clinical assessment. Similarly, for IAVB, PR interval duration was identified as a key feature, aligning with established diagnostic criteria. However, the features associated with LBBB and RBBB present an interesting case. A characteristic feature of RBBB is discordant T-waves in Lead 7 and potential T-wave inversion in Leads 7 to 9. Our

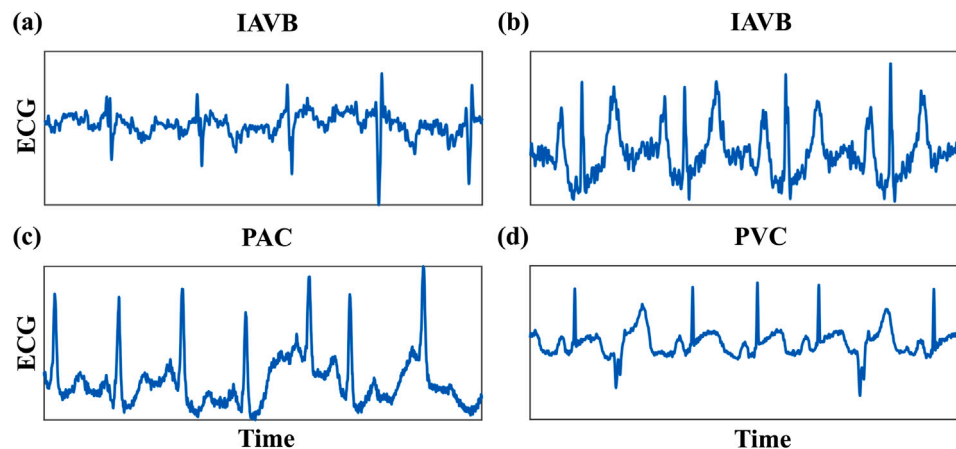


Fig. 8. Sample Lead 2 ECG segments for (a) IAVB, (b) IAVB, (c) PAC, and (d) PVC.

model identified the top-performing features associated with RBBB as the positive and negative areas between the ECG signal and a line segment connecting the S and T peaks in Lead 7. These features, which demonstrate abnormal values in the presence of discordant T-waves, indirectly indicate the presence of RBBB and emphasize the model's ability to capture subtle yet clinically relevant measurements. Similarly, the characteristic clinical feature associated with LBBB is a widened QRS complex exceeding 120 ms. Our model identified the spectral spread of the median beat and the width of the temporal wavelet transform projection as key features for LBBB classification, highlighting the effectiveness of frequency and time-frequency representations in distinguishing LBBB. The temporal wavelet transform projection width, which is associated with cardiac electrical activity duration, is closely correlated with QRS width, as an increase in QRS width typically results in a corresponding increase in electrical activity duration. Interestingly, the model highlighted the temporal wavelet transform projection width as a key feature instead of the QRS width itself. This suggests that it may have captured novel aspects of LBBB, such as the importance of total electrical activity within the cardiac cycle. This aligns with established clinical observations while offering deeper insights into the complexities of this arrhythmia. Similarly, in the case of PAC, RR interval volatility was identified as the most important model feature. However, the current feature set exhibited limitations in characterizing PVC, STD and STE. In the case of PVC, the standard deviation of each beat's correlation with the median beat partially captured the irregularity of ectopic beats compared to other beats. However, the model failed to identify the primary cause, as no features specifically corresponding to abnormal beats were highlighted. For STD, RR interval volatility was identified as the most dominant feature, and for STE, the TP interval was identified as the most dominant feature. Both of these diverge from typical clinical diagnostic criteria and do not align with standard clinical assessments. Despite these limitations, the model successfully recognized the importance of the ST segment as the second most dominant feature for both STD and STE. These findings, along with the previously observed low performance in STE detection, highlight the challenges associated with accurately identifying ST-segment changes due to subjective interpretation and potential inconsistencies in ECG labeling. Overall, these results illustrate that the proposed feature set significantly enhances the interpretability of our model and validates its capability for providing meaningful insights into cardiac arrhythmia diagnoses clinically.

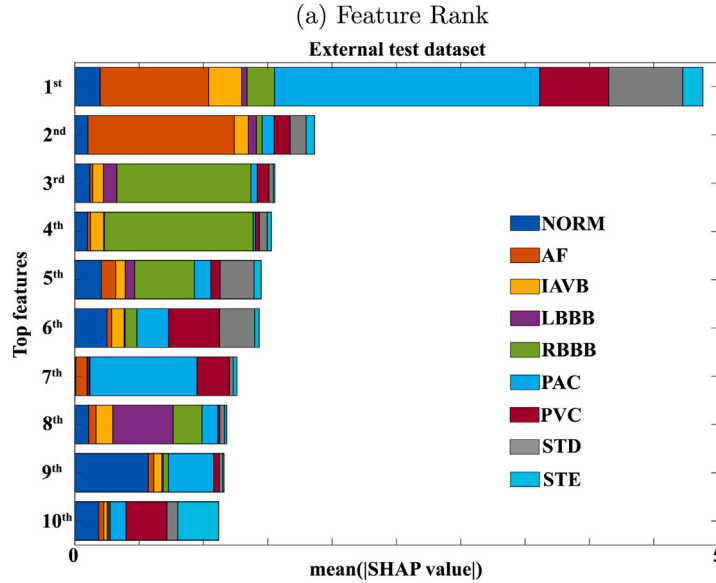
Finally, we investigated the features contributing most to the model's overall performance on the external test dataset. Fig. 9 illustrates the top ten features and their corresponding SHAP value distributions across all classes. In general, the mean SHAP value for rhythm, morphology, time-frequency, and miscellaneous features was found to be

0.084, 0.072, 0.092, and 0.089, respectively. Hence, time-frequency-based features demonstrated the highest efficiency, surpassing the next best feature category by 34%. Although morphology-based features exhibited lower individual efficiency, their collective contribution to overall performance was substantial due to the large number of features. RR interval volatility was identified as the top feature, with a SHAP values exceeding those of the second important feature by 162%. This feature played a significant role in identifying AF, PAC, and PVC, accounting for 17%, 42%, and 11%, respectively, of its SHAP value. This highlights the feature's effectiveness in capturing the irregular rhythmic nature of these arrhythmias. The commonly used clinical metric, heart rate variability, was identified as the second most important feature. This feature was primarily associated with AF identification, contributing 61% of its SHAP value to this class. The next two top features were the positive and negative areas between the Lead 7 ECG segment and the S-T peak line segment. These were primarily associated with RBBB identification, contributing 67% and 76% of their respective SHAP values to this class. The seventh most important feature, representing the abnormal beat, was predominantly associated with PAC, as expected, accounting for 66% of the SHAP values. The eighth most important feature, identified as the most significant feature for LBBB, was the spectral spread of the median beat, with LBBB contributing to 40% of its SHAP value. Overall, these results demonstrate that the proposed feature set significantly enhances the model's interpretability and provides valuable insights into cardiac arrhythmia diagnoses.

4.4. Limitations and future work

Several limitations of the proposed arrhythmia detection approach using interpretable features exist. A primary challenge in feature-based ML modeling is the model's performance dependency on the effectiveness of the extracted features. Inefficient feature representation of the arrhythmia set can significantly impact model performance. Since the proposed method involves a step-by-step process of denoising and morphology detection prior to feature extraction, errors introduced during these stages can propagate and affect model performance. Specifically, using Lead 2 as the basis for ECG morphology detection for all 12 leads, rather than performing independent morphology detection, limits the extracted features' effectiveness in representing the cardiac activity captured by other leads. Additionally, XGBoost, while effective for many tasks, is inherently designed for tabular data and may not fully capture the temporal dynamics present in ECG signals. The reliance on statistical summaries of temporal features, such as the use of median beats, can lead to a loss of information regarding waveform dependencies. Another significant challenge in automated arrhythmia detection is the inherent imbalance present in available

Feature rank	Feature
1st	RR interval volatility
2nd	Heart rate variability
3rd	Positive area: S-T peak line vs. ECG segment (lead 7)
4th	Negative area: S-T peak line vs. ECG segment (lead 7)
5th	Median PR interval (lead 7)
6th	Peak magnitude of temporal projected median WT (lead 1)
7th	Median TP interval of abnormal RR int (type 1) (lead 4)
8th	Spectral spread of median beat (lead 8)
9th	SD of TP interval (lead 11)
10th	Mean zero-lag correlation with median beat (lead 8)



(b) SHAP value distribution for top 10 features

Fig. 9. Top feature to the overall performance of the model and its SHAP value distribution.

datasets. The datasets utilized in this study often exhibited imbalanced distributions of arrhythmia classes, which can hinder the performance of XGBoost models and lead to biased predictions. Furthermore, the effectiveness of the proposed method heavily relies on accurate and consistent labeling. Given the complex and nuanced nature of ECG interpretation, which often results in substantial inter-cardiologist variability in diagnosis, generalizing the proposed method will require training on multiple datasets which will be explored in future work. Additionally, the performance comparison of the proposed method with existing literature in this study is constrained by the lack of publicly available implementation details, including code, sampling strategies, and random seed settings. Moreover, as evident from the multi-label nature of existing ECG databases, multiple abnormalities frequently co-exist within a single recording. The proposed study considered a limited number of arrhythmias individually, limiting its ability to represent complex cases involving multiple simultaneous arrhythmias.

Future work should aim to mitigate these limitations by improving denoising and morphology detection techniques to reduce erroneous feature extraction and enhance performance. Specifically, implementing morphology detection independently across all leads, rather than relying predominantly on Lead 2, could improve the model's robustness and accuracy. Expanding the feature set with advanced signal processing tools, such as discrete cosine transform and model analysis methods such as dynamic mode decomposition, could further capture subtle differences between arrhythmias, providing deeper insights and improving classification accuracy. Furthermore, future efforts will explore

the efficacy of the proposed features in representing and classifying simultaneous arrhythmias.

5. Conclusions

This study proposes a novel approach for arrhythmia detection using handcrafted interpretable features extracted from 12-lead ECG signals. A diverse set of 654 features, spanning 60 feature types, was engineered using mathematical techniques such as the Fourier transform, wavelet transform, and cross-correlation. These features were incorporated into an XGBoost model framework, which was optimized through 10-fold cross-validation and hyperparameter tuning. The model was then trimmed to the top performing 159 features based on the SHAP framework. The efficacy of the resulting feature subset was then evaluated on an external test dataset comprising data from multiple sources. To understand the model's decision-making process, SHAP value analysis was conducted. The proposed approach achieved comparable accuracy to deep learning models while offering enhanced interpretability, training efficiency, and adaptability. Notably, the proposed method exhibited a 4% performance reduction during 10-fold cross-validation compared to the CNN model by Chen et al. [17], but achieved a 48% improvement compared to the CNN model and a 2% improvement compared to the multi-stage DL model by Lai et al. [16] when tested on the external dataset. Overall, the proposed method demonstrated strong performance in detecting AF and block abnormalities (IAVB, LBBB and RBBB). Further investigation into feature importance using extracted

SHAP values revealed that the proposed method aligns with standard 12-lead ECG interpretation practices. Key features such as heart rate variability, PR interval, QRS complex width, and abnormal beats were found to be crucial for identifying AF, IAVB, LBBB, and PAC, respectively. Thus, the proposed method offers enhanced interpretability, providing valuable insights into arrhythmia diagnoses and supporting clinical utility.

CRedit authorship contribution statement

Jibin Joy Kolliyl: Writing – original draft, Methodology, Investigation, Formal analysis. **Melissa C. Brindise:** Writing – review & editing, Supervision, Project administration, Investigation, Conceptualization.

Ethics statement

No ethics statement is required for this work. This research made use explicitly of publicly available data. We conducted no human subjects experiments nor independently collected any clinical or human subjects data.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT and Gemini in order to improve the clarity, readability, and language of this manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset used in this study, along with the code used for feature extraction, data sampling, and model development, is available on the following GitLab repository: <https://git.psu.edu/brindise-lab/multiclass-arrhythmia-detection>.

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